Stability Time from an Underdamped Second-Order Model

From `gnn.png` and `dM_dt.png` to a practical value of t_{stable}

$$t_s \approx 2\,\mathrm{s}, \qquad \tau \approx 2\,\mathrm{s}, \qquad t_{stable} \approx 10\,\mathrm{s}.$$

1. From `gnn.png` to the model

The curve in `gnn.png` has four features:

$$\text{rise} \;\rightarrow\; \text{overshoot} \;\rightarrow\; \text{decaying oscillation} \;\rightarrow\; \text{final value}.$$

This is the standard signature of an **linear underdamped second-order system**.

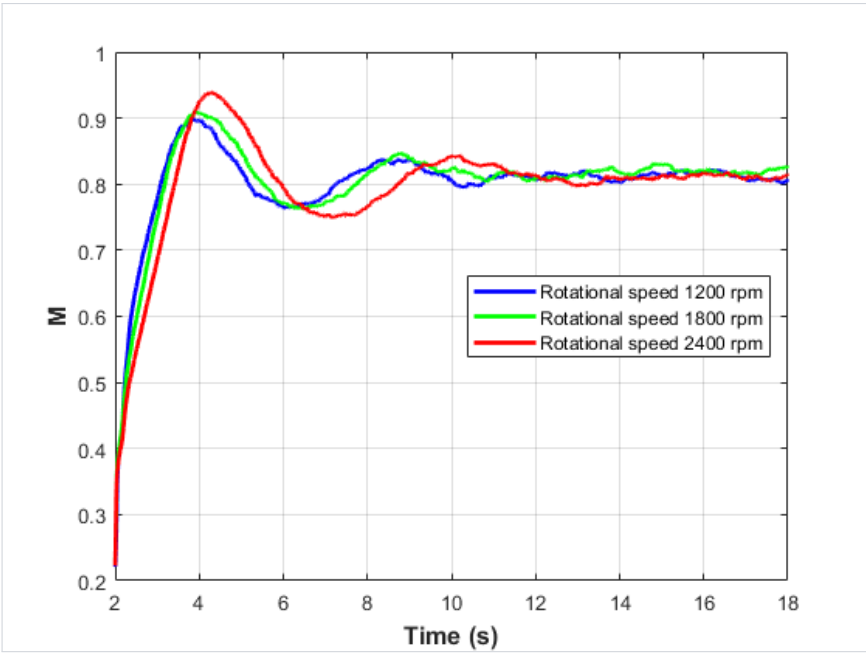

Figure 1: Observed response used to motivate the underdamped second-order model.

We model it by

$$\ddot{M} + 2\zeta\omega_n\dot{M} + \omega_n^2(M - M_\infty) = 0, \qquad t \geq t_s.$$

Let

$$u = t - t_s, \qquad \tau = \frac{1}{\zeta\omega_n}, \qquad \omega_d = \omega_n\sqrt{1-\zeta^2}.$$

Then the underdamped response has the form

$$M(t) = M_\infty + e^{-u/\tau}\left(A\cos(\omega_d u) + B\sin(\omega_d u)\right).$$

For the step-like case with

$$M(t_s) = M_0, \qquad \dot{M}(t_s) = 0, \qquad \Delta M = M_\infty - M_0,$$

we obtain

$$M(t) = M_\infty - \Delta M\, e^{-u/\tau} \left[\cos(\omega_d u) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d u) \right].$$

2. Stability Time from the model

The transient error is

$$e(t) = M(t) - M_\infty.$$

Its envelope decays as

$$|e(t)| \propto e^{-u/\tau}.$$

Also,

$$\left| \frac{dM}{dt} \right| \propto e^{-u/\tau}.$$

Therefore a compact practical rule is

$$t_{stable} \approx t_s + 4\tau.$$

Useful variants:

$$t_s + 3\tau : \text{ loose}, \qquad t_s + 4\tau : \text{ recommended}, \qquad t_s + 5\tau : \text{ conservative}.$$

3. Estimate τ from the two plots

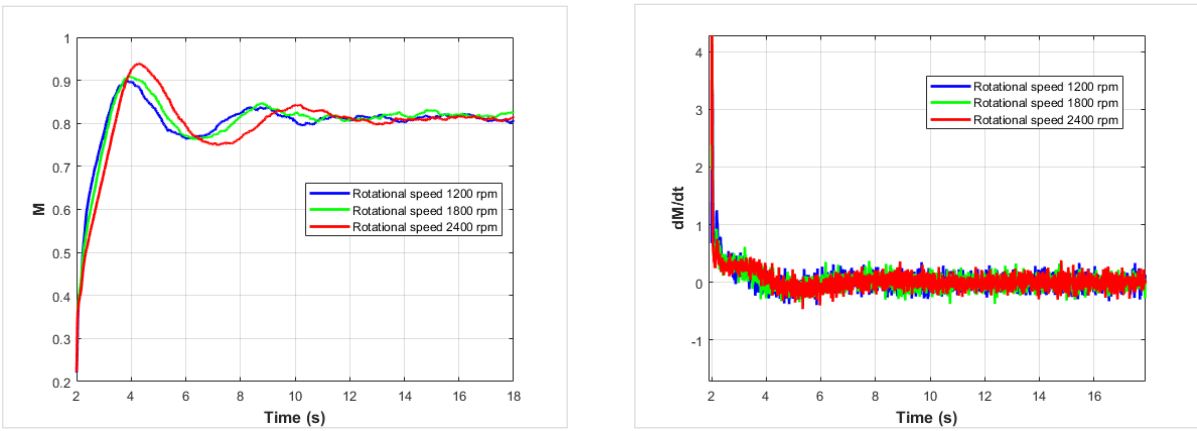

Figure 2: Left: $M(t)$. Right: dM/dt.

From `gnn.png`, the transient starts at about

$$t_s \approx 2\,\text{s}.$$

From `dM_dt.png`, the derivative envelope drops from order

$$|\dot{M}| \sim 4 \quad \text{near } t \approx 2\,\text{s}$$

to order

$$|\dot{M}| \sim 0.08 \quad \text{near } t \approx 10\,\text{s}.$$

Hence

$$\frac{0.08}{4} \approx 0.02 \approx e^{-4}.$$

Using the decay law

$$|\dot{M}(t)| \propto e^{-(t-t_s)/\tau},$$

we get

$$e^{-(10-2)/\tau} \approx e^{-4} \quad \Rightarrow \quad \tau \approx 2\,\text{s}.$$

4. Numerical value of t_{stable}

Substitute the estimated τ into the recommended rule:

$$t_{stable} \approx t_s + 4\tau \approx 2 + 4 \times 2 = 10\,\text{s}.$$

If a safer margin is needed:

$$t_{stable} \approx t_s + 5\tau \approx 2 + 5 \times 2 = 12\,\text{s}.$$

$$\boxed{t_{stable} \approx 10\,\text{s}} \qquad \boxed{t_{stable} \approx 12\,\text{s (conservative)}}.$$